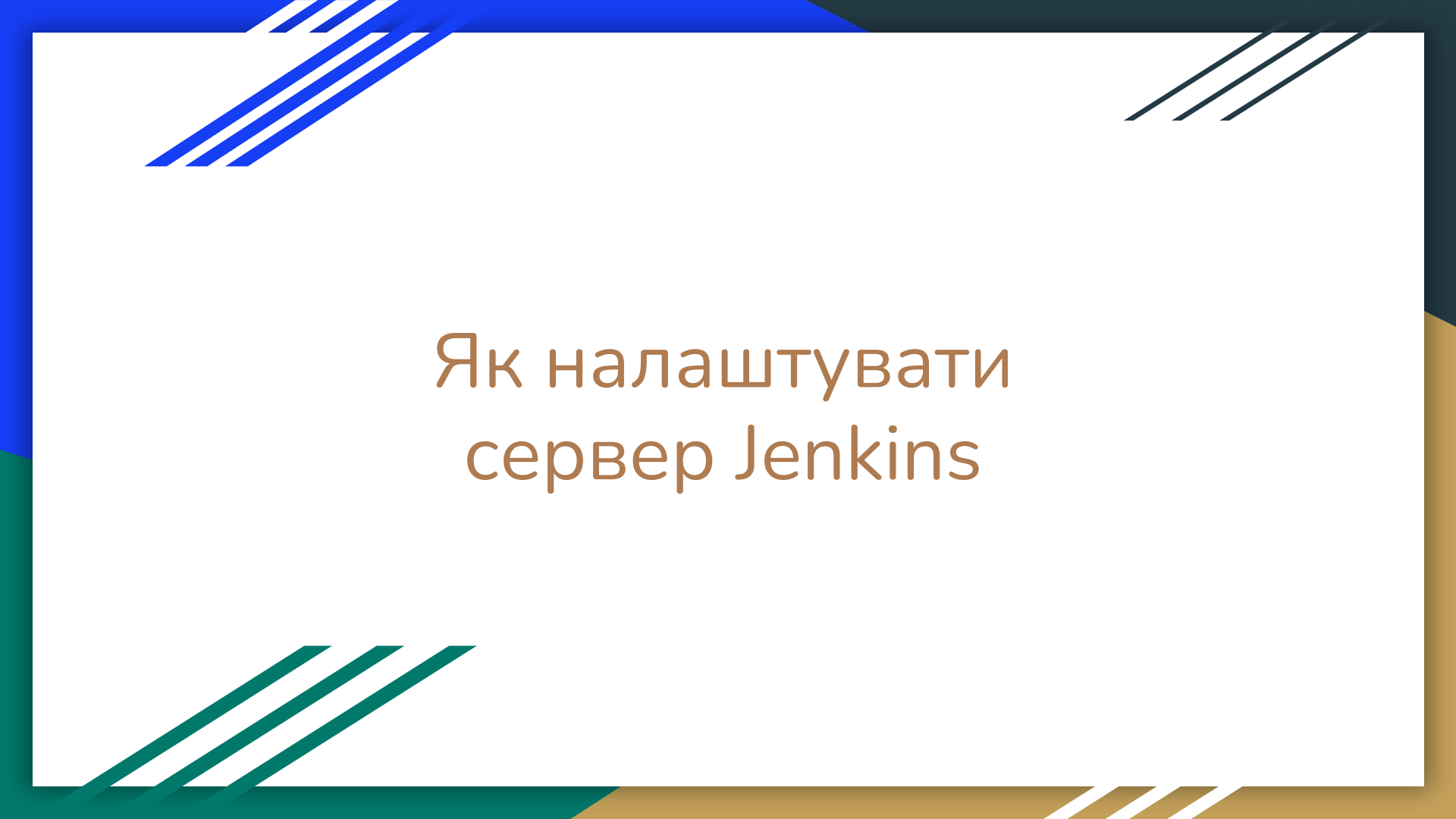




Як налаштувати сервер Jenkins

Що таке Jenkins

Jenkins

Build great things at any scale

The leading open source automation server, Jenkins provides hundreds of plugins to support building, deploying and automating any project.

Велика ІДЕЯ

Зробіть щось автоматично (без рук), що передбачає:

- Кілька кроків
- Кроки мають залежності
- Можливо, доведеться виконувати кілька завдань на кількох серверах

Один із прикладів того, що ви можете зробити з Jenkins

- О 3:21 у неділю
- Отримайте маркер зі сховища за допомогою попередньо визначених змінних автентифікації (вставлених OPS)
- Увійдіть у Google Cloud за допомогою цього маркера
- Скопіюйте 2 ТБ даних із HDFS
- Обробіть ці дані за допомогою програми Python
- Скопіюйте вихідні дані в Google Dataflow
- Якщо це не вдається або займає більше 60 хвилин, перейдіть за викликом за допомогою плагіна Pagerduty
- Якщо це вдасться, надішліть нашій команді електронний лист.
- Видаліть маркер, оскільки безпека не дозволяє маркерам зберігатися поза Сейфом.
- Зберігайте докладні журнали для 30 останніх запусків.
- Також зберігайте інші метадані, як-от час виконання.

Інший приклад

- Візьміть нову версію коду з git repo
- Інтегруйте його з maven або іншим інструментом
- Розгорніть його на 500 серверах
- Проведіть автоматичне тестування
- Визначив, чи хороша ця нова версія чи ні.
- Якщо автоматичне тестування зазнає невдачі, відкотіть усе назад і перейдіть за викликом

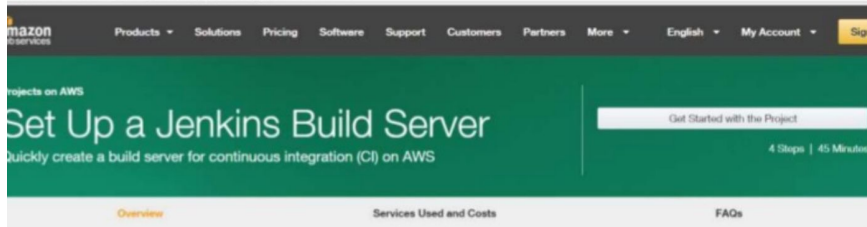
Існують інші подібні інструменти

	Jenkins	Bamboo	Travis
Open Source	Yes	No	Yes
Plugins	Yes (Lots of them)	Expensive	Few
Support	Yes (Being Open Source helps)	Not Much (Small user community)	No
Easy Usage	No (The plugins help)	Yes	Yes
Atlassian Product Integration	Yes	Yes	Yes
Repo Support	All	All	Github

Ви можете використовувати Jenkins на AWS

(якщо вам не хочеться керувати своїми)

<https://aws.amazon.com/getting-started/projects/setup-jenkins-build-server/>



In this project, you will learn how to deploy and host Jenkins, an open-source automation software predominantly used for CI/CD (Continuous Integration/Continuous Deployment). You will use Amazon Elastic Compute Cloud (EC2) in a public subnet within your own Amazon Virtual Private Cloud (VPC) and you will also set up an Amazon Elastic Block Store (EBS) volume.

What you'll accomplish:

Launch a virtual application server to host your Jenkins installation using Amazon EC2. In this single-node setup, both the Jenkins server and agent run on the same Amazon EC2 instance. The instance type you provision will match your expected load. You will also be able to resize the instance if your expected load changes.

Deploy Jenkins on Amazon EC2. You will download and deploy the latest version of Jenkins.

Configure Jenkins with build servers. You will configure Jenkins so that it spins up additional Amazon EC2



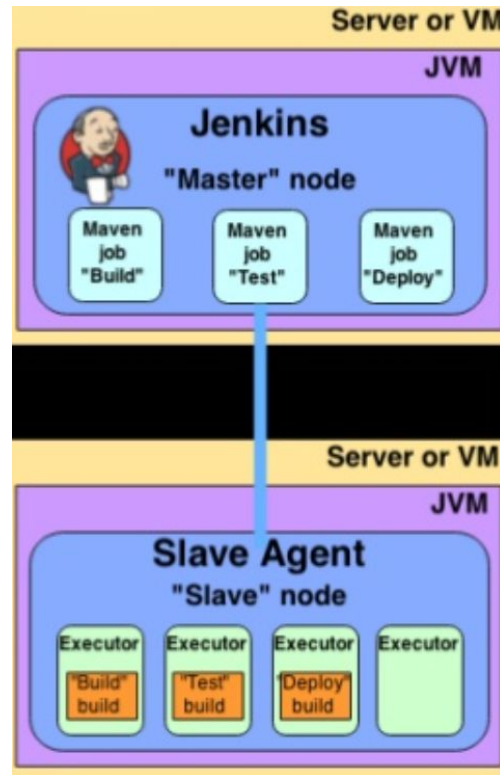
Встановлення

Java, Jenkins



Архітектура Jenkins

- Ви можете мати багато підлеглих вузлів
- Кожен підлеглий вузол може мати мітки.
- На підлеглих вузлах не потрібно нічого, окрім Java.
- Лише jar-файл, який основний вузол передає під час встановлення агента



Увійдіть у Jenkins через улюблений браузер



The image shows the Jenkins login page. At the top, there is a black header bar with the Jenkins logo (a cartoon character) on the left, the word "Jenkins" in white in the center, and a "log In" link in white on the right. Below the header, there is a light green horizontal bar with the word "Jenkins" and a right-pointing arrow. The main content area is white and contains the login form. It includes labels "User:" and "Password:" followed by text input fields. Below the password field is a checkbox labeled "Remember me on this computer". At the bottom of the form is a blue button with the text "log In".

Jenkins

log In

Jenkins

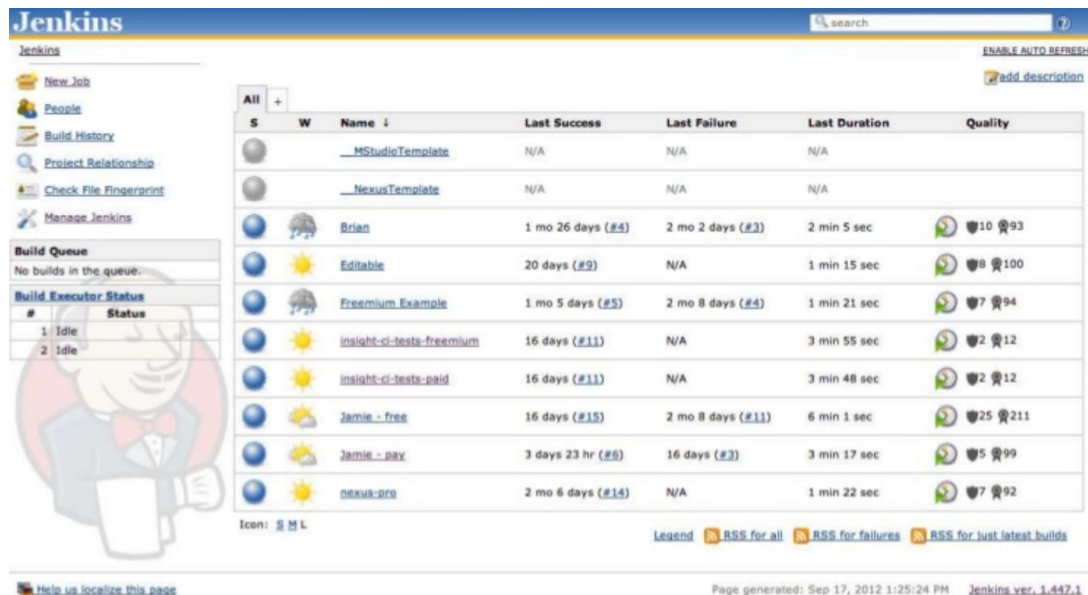
User:

Password:

☐ Remember me on this computer

log In

Перша сторінка, яку ви бачите (ваш досвід може відрізнятися)



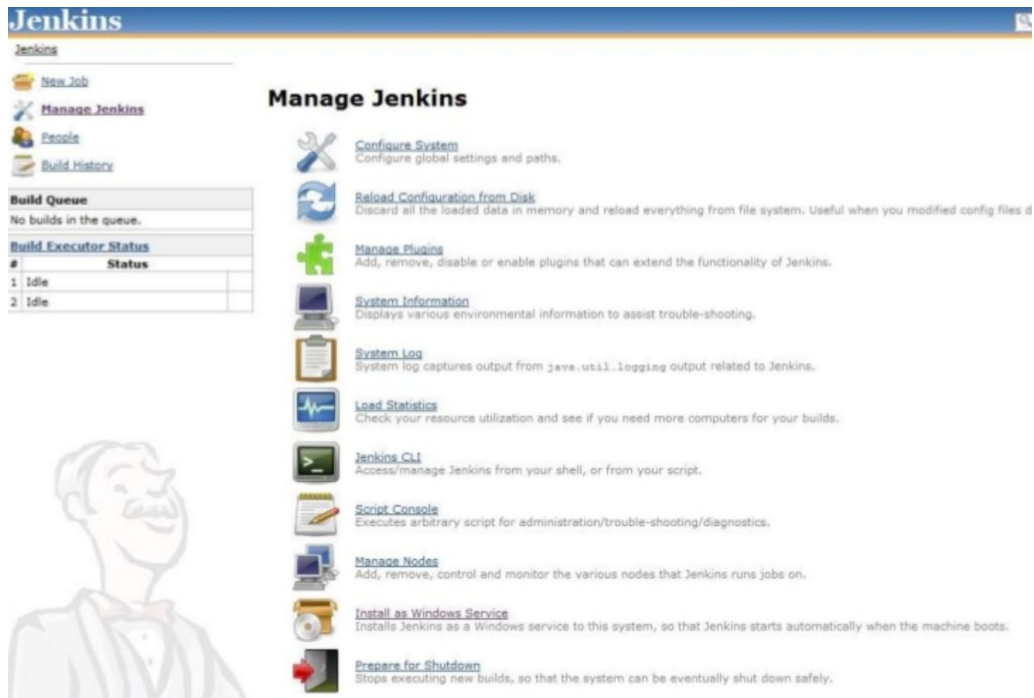
The screenshot displays the Jenkins web interface. On the left, there is a sidebar with navigation links: [New Job](#), [People](#), [Build History](#), [Project Relationship](#), [Check File Fingerprint](#), and [Manage Jenkins](#). Below these links, the **Build Queue** section shows "No builds in the queue." The **Build Executor Status** section shows two executors, both in an "Idle" state.

The main content area features a table of jobs. The table has columns for **S** (Status), **W** (Icon), **Name**, **Last Success**, **Last Failure**, **Last Duration**, and **Quality**. The jobs listed are:

S	W	Name	Last Success	Last Failure	Last Duration	Quality
		_MStudioTemplate	N/A	N/A	N/A	
		_NexusTemplate	N/A	N/A	N/A	
		Brian	1 mo 26 days (#4)	2 mo 2 days (#3)	2 min 5 sec	10 93
		Editable	20 days (#9)	N/A	1 min 15 sec	8 100
		Freemium Example	1 mo 5 days (#5)	2 mo 8 days (#4)	1 min 21 sec	7 94
		insight-ci-tests-freemium	16 days (#11)	N/A	3 min 55 sec	2 12
		insight-ci-tests-paid	16 days (#11)	N/A	3 min 48 sec	2 12
		Jamie - free	16 days (#15)	2 mo 8 days (#11)	6 min 1 sec	25 211
		Jamie - pay	3 days 23 hr (#6)	16 days (#3)	3 min 17 sec	5 99
		nexus-pro	2 mo 6 days (#14)	N/A	1 min 22 sec	7 92

At the bottom of the table, there is a **Legend** section with links for [RSS for all](#), [RSS for failures](#), and [RSS for just latest builds](#). The footer of the page indicates "Page generated: Sep 17, 2012 1:25:24 PM" and "Jenkins ver. 1.447.1".

Керування екраном Jenkins



The screenshot displays the Jenkins web interface. On the left, a sidebar contains navigation links: 'New Job', 'Manage Jenkins', 'People', and 'Build History'. Below these is a 'Build Queue' section showing 'No builds in the queue.' and a 'Build Executor Status' table with two idle executors. The main area is titled 'Manage Jenkins' and lists various administrative tasks, each with an icon and a brief description. A faint cartoon character is visible in the bottom-left corner of the interface.

Jenkins

[New Job](#)

[Manage Jenkins](#)

[People](#)

[Build History](#)

Build Queue

No builds in the queue.

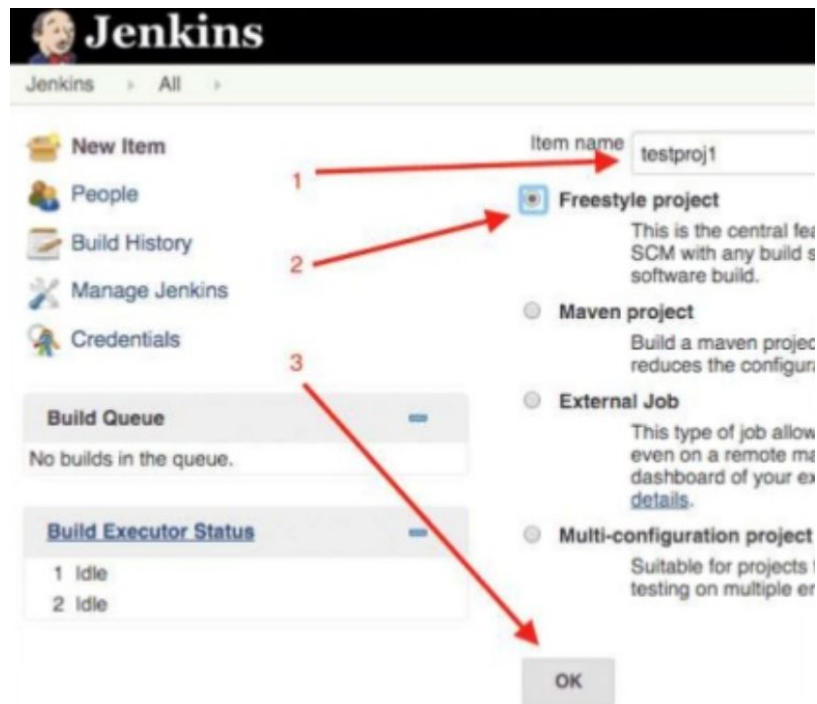
Build Executor Status

#	Status
1	Idle
2	Idle

Manage Jenkins

-  [Configure System](#)
Configure global settings and paths.
-  [Reload Configuration from Disk](#)
Discard all the loaded data in memory and reload everything from file system. Useful when you modified config files d
-  [Manage Plugins](#)
Add, remove, disable or enable plugins that can extend the functionality of Jenkins.
-  [System Information](#)
Displays various environmental information to assist trouble-shooting.
-  [System Log](#)
System log captures output from `java.util.logging` output related to Jenkins.
-  [Load Statistics](#)
Check your resource utilization and see if you need more computers for your builds.
-  [Jenkins CLI](#)
Access/manage Jenkins from your shell, or from your script.
-  [Script Console](#)
Executes arbitrary script for administration/trouble-shooting/diagnostics.
-  [Manage Nodes](#)
Add, remove, control and monitor the various nodes that Jenkins runs jobs on.
-  [Install as Windows Service](#)
Installs Jenkins as a Windows service to this system, so that Jenkins starts automatically when the machine boots.
-  [Prepare for Shutdown](#)
Stops executing new builds, so that the system can be eventually shut down safely.

Створення нової роботи (проект)



Створення нової роботи

The screenshot shows the Jenkins web interface for configuring a new project. The top navigation bar includes the Jenkins logo, a search bar, and user links for 'u11' and 'log out'. The breadcrumb trail indicates the current path: 'Jenkins > MySweetAppCI > configuration'. On the left sidebar, there are links for 'Back to Dashboard', 'Status', 'Changes', 'Workspaces', 'Build Now', 'Delete Project', and 'Configure'. Below these is a 'Build History' section with a '(trend)' link and two RSS feeds: 'RSS for all' and 'RSS for failures'. The main configuration area is titled 'MySweetAppCI configuration'. It contains a 'Project name' field with the value 'MySweetAppCI'. The 'Description' field contains the text 'Builds MySweetApp to ensure I didn't break anything or forget a file.' Below the description is a 'Preview' link. There are three checkboxes: 'Discard Old Builds' (unchecked), 'This build is parameterized' (unchecked), and 'Disable Build (No new builds will be executed until the project is re-enabled.)' (unchecked). A 'Github project' field contains the value 'git@github.com:mygithubusername/MySweetApp.git/'. Below these are two more checkboxes: 'Execute concurrent builds if necessary' (unchecked). A section titled 'Advanced Project Options' has an 'Advanced...' button. Under 'Source Code Management', there are three radio buttons: 'CVS' (unchecked), 'CVS Projectset' (unchecked), and 'Git' (checked). The 'Repositories' section shows a 'Repository URL' field with the value 'git@github.com:mygithubusername/MySweetApp.git/'.

Jenkins

MySweetAppCI configuration

Project name: MySweetAppCI

Description: Builds MySweetApp to ensure I didn't break anything or forget a file.

Preview

☐ Discard Old Builds

Github project: git@github.com:mygithubusername/MySweetApp.git/

☐ This build is parameterized

☐ Disable Build (No new builds will be executed until the project is re-enabled.)

☐ Execute concurrent builds if necessary

Advanced Project Options

Advanced...

Source Code Management

☐ CVS

☐ CVS Projectset

☒ Git

Repositories

Repository URL: git@github.com:mygithubusername/MySweetApp.git/

Створення нової роботи

General **Source Code Management** Build Trigger Build Environment Bindings Build Post-build Actions

☐ None
☒ Git

Repositories

Repository URL

Credentials

Branches to build

Branch Specifier (blank for 'any')

Repository browser

Additional Behaviours

☐ Clean before checkout

Build a New Job (2)

Користувачі

☒ Enable project-based security

User/group	Job							Run		SCM
	Delete	Configure	Read	Discover	Build	Workspace	Cancel	Delete	Update	Tag
 xxx	☐	☐	☒	☐	☒	☐	☐	☐	☐	☐
 zzz	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒
Anonymous	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

User/group to add:

Add

Ролі

 **Manage Roles**

Global roles

Role	Overall			Slave		Job					Run			View			SCM
	Administrator	Read	Configure	Delete	Create	Delete	Configure	Read	Build	Workspace	Delete	Update	Artifacts	Create	Delete	Configure	Tag
admin	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒
anonymous	☐	☒	☐	☐	☐	☐	☐	☒	☐	☐	☐	☐	☐	☐	☐	☐	☐
job-creator	☐	☒	☐	☐	☒	☐	☐	☒	☐	☐	☐	☐	☐	☐	☐	☐	☐

Role to add

Add

Project roles

Role	Pattern	Job					Run		
		Delete	Configure	Read	Build	Workspace	Delete	Update	Artifacts
		☐	☒	☒	☒	☒	☐	☒	☒
		☒	☒	☒	☒	☒	☒	☒	☒
		☐	☒	☒	☒	☒	☐	☒	☒
Sample Sample.*		☐	☒	☒	☒	☒	☐	☒	☒
		☐	☒	☒	☒	☒	☐	☒	☒

Role to add

Pattern

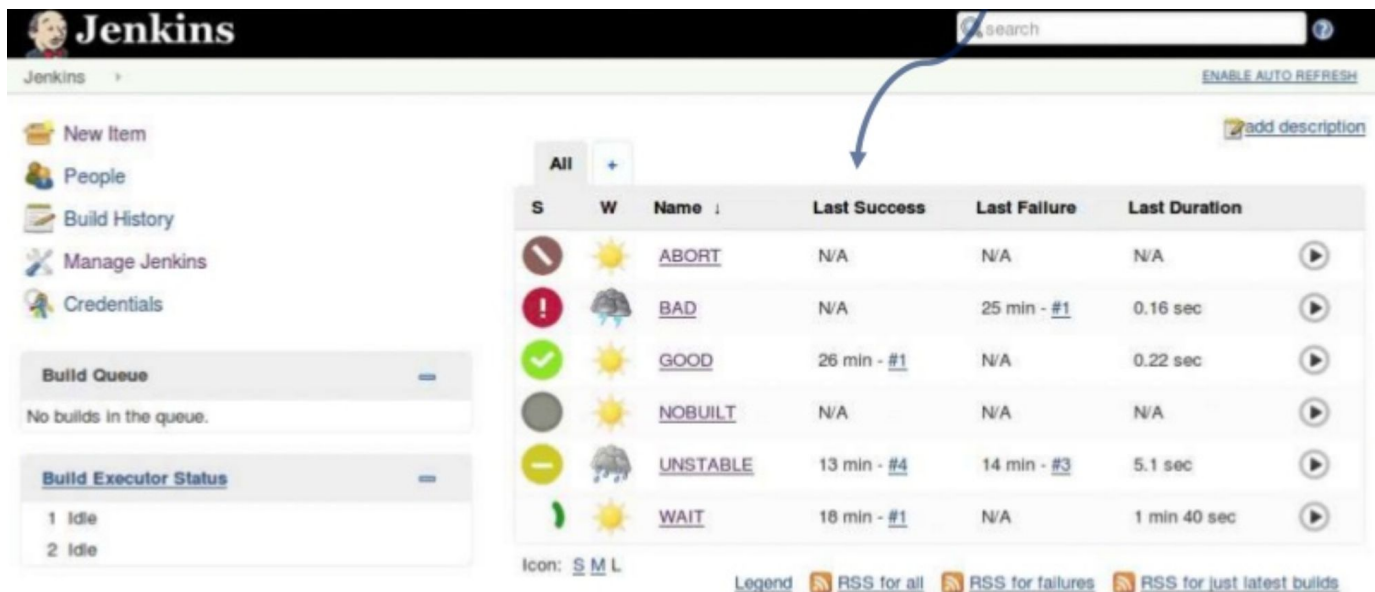
Add

Save

СТВОРЕННЯ ІСТОРІЇ

[illegible]

Підсумок статусу кожного повторюваного завдання



The screenshot shows the Jenkins dashboard. On the left, there is a sidebar with navigation links: New Item, People, Build History, Manage Jenkins, and Credentials. Below these are two expandable sections: 'Build Queue' (showing 'No builds in the queue.') and 'Build Executor Status' (showing '1 Idle' and '2 Idle'). The main content area displays a table of build statuses. A blue arrow points from the search bar at the top right to the table. The table has columns for status icons (S and W), build names, last success, last failure, and last duration. The status icons are color-coded: green for success, red for failure, grey for no build, yellow for unstable, and green with a lightning bolt for wait. The table lists six build statuses: ABORT, BAD, GOOD, NOBUILT, UNSTABLE, and WAIT. Below the table, there is a legend for the status icons and links for RSS feeds.

S	W	Name	Last Success	Last Failure	Last Duration
		ABORT	N/A	N/A	N/A
		BAD	N/A	25 min - #1	0.16 sec
		GOOD	26 min - #1	N/A	0.22 sec
		NOBUILT	N/A	N/A	N/A
		UNSTABLE	13 min - #4	14 min - #3	5.1 sec
		WAIT	18 min - #1	N/A	1 min 40 sec

Icon: [S](#) [M](#) [L](#)

Legend [RSS for all](#) [RSS for failures](#) [RSS for just latest builds](#)

Глобальна конфігурація інструменту

Jenkins > Global Tool Configuration

[Back to Dashboard](#)

[Manage Jenkins](#)

Global Tool Configuration

Maven Configuration

Default settings provider:

Default global settings provider:

JDK

JDK installations:

List of JDK installations on this system

Git

Git installations

Git	
Name	Default
Path to Git executable	git
☐ Install automatically	

Скільки є плагінів?



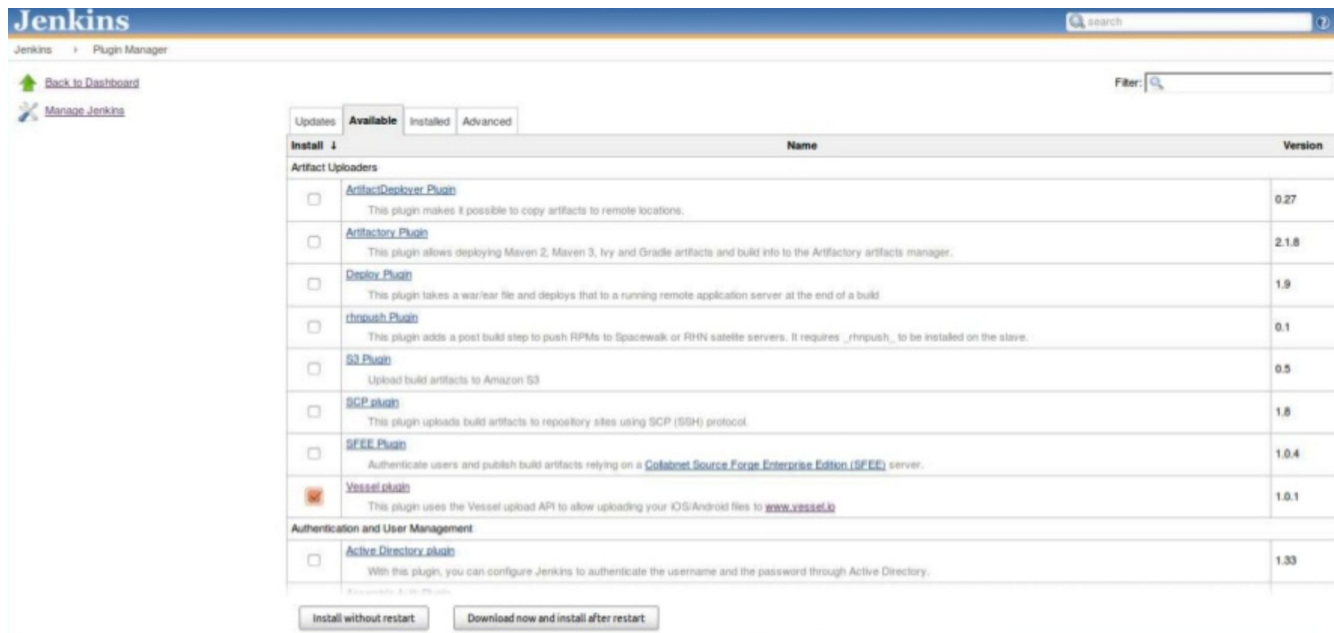
Plugins Index

Discover the 1000+ community contributed Jenkins plugins to support building, deploying and automating any project.

Browse Find plugins... 🔍

Browse categories	New Plugins	Recently updated	Trending
Platforms	Autocomplete Parameter	Delivery Pipeline	Pipeline: Stage Tags Metadata
User interface	Spring Initializr	Cisco Spark Notifier	Pipeline: GitHub Groovy Libraries
Administration	DotNet as Script	Docker Pipeline	Pipeline: Model API
Source code management	FlexDeploy	Last Changes	Pipeline: Model Definition
Build management	SWAMP	ElasticBox CI	Docker Pipeline
	WSO2 Oauth	bootstrapped-multi-test-results-report	Authentication Tokens API
	Display Console Output	GitHub Pull Request Coverage Status	Docker Commons
	Block Same build	Pipeline Maven Integration	Display URL API
	FTP Rename	JGiven	Credentials Binding

Керування плагінами



The screenshot displays the Jenkins Plugin Manager interface. At the top, the Jenkins logo and a search bar are visible. Below the header, there are links for 'Back to Dashboard' and 'Manage Jenkins'. The main section is titled 'Available' and contains a table of plugins. The table has columns for 'Name' and 'Version'. The plugins listed are categorized under 'Artifact Uploaders' and 'Authentication and User Management'. At the bottom, there are two buttons: 'Install without restart' and 'Download now and install after restart'.

Available	
Name	Version
Artifact Uploaders	
☐ ArtifactDeployer Plugin This plugin makes it possible to copy artifacts to remote locations.	0.27
☐ Artifactory Plugin This plugin allows deploying Maven 2, Maven 3, Ivy and Gradle artifacts and build info to the Artifactory artifacts manager.	2.1.8
☐ Deploy Plugin This plugin takes a war/ear file and deploys that to a running remote application server at the end of a build.	1.9
☐ rhpush Plugin This plugin adds a post build step to push RPMs to Spacewalk or RHN satellite servers. It requires 'rhpush' to be installed on the slave.	0.1
☐ S3 Plugin Upload build artifacts to Amazon S3	0.5
☐ SCP plugin This plugin uploads build artifacts to repository sites using SCP (SSH) protocol.	1.8
☐ SFE Plugin Authenticate users and publish build artifacts relying on a Colabnet Source Forge Enterprise Edition (SFE) server.	1.0.4
☒ Vessel plugin This plugin uses the Vessel upload API to allow uploading your iOS/Android files to www.vessel.io	1.0.1
Authentication and User Management	
☐ Active Directory plugin With this plugin, you can configure Jenkins to authenticate the username and the password through Active Directory.	1.33

[Install without restart](#) [Download now and install after restart](#)

Перегляди

The screenshot shows the Jenkins web interface. On the left is a sidebar with navigation links: New Job, People, Build History, Edit View, Delete View, Manage Jenkins, and UI Samples. Below these are sections for 'Build Queue' (showing no builds) and 'Build Executor Status' (showing two idle executors). The main area is titled 'Recently Chosen Views' and contains a table of views. A dropdown menu is open over the 'Important View' tab, showing a search bar with 'V' and a list of view options: 'Beautiful View' (highlighted), 'Important View', 'Nice View', and 'View of Project X'. A tooltip for 'View of Project X' is visible, stating 'This view has a longer name and several spaces'.

Jenkins search ?

Jenkins Important View ENABLE AUTO REFRESH add description

Recently Chosen Views

All Generation Z **Important View** Nice View View of Project X

S	W	Name ↓	Last Success
		8.02.280-1.2-patch	N/A
		8.02.280P-1.2-patch	N/A
		8.02.281-1.2-patch	N/A
		8.02.281P-1.2-patch	N/A
		8.02.307-1.2-patch	N/A
		8.02.308-1.2-patch	N/A
		8.02.351-1.2-patch	N/A
		8.02.352-1.2-patch	N/A
		8.02.352P-1.2-patch	N/A

Choose another view +

V| search

Choose another view

Beautiful View

Important View

Nice View

This view has a longer name and several spaces

View of Project X

Build Queue

No builds in the queue.

Build Executor Status

#	Status
1	Idle
2	Idle

Logger Console Expand

Встановіть підлеглий агент на нову віртуальну машину

Name	QA
Description	Jenkins slave for deployments
# of executors	1
Remote root directory	C:\Jenkins\
Labels	qa
Usage	Only build jobs with label restrictions matching this node
Launch method	Launch slave agents via Java Web Start
Tunnel connection through	
JVM options	
Availability	Keep this slave on-line as much as possible

Node Properties

- ☐ Environment variables
- ☐ Tool Locations

Save

Іноді це трапляється після перезавантаження підлеглого вузла



Slave Test Slave Node

Mark this node temporarily offline

 This node is offline because Jenkins failed to launch the slave agent on it. [See log for more details](#)

Connect slave to Jenkins one of these ways:

-  Launch Launch agent from browser on slave
- Run from slave command line:

```
javaws http://localhost:8080/computer/Test%20Slave%20Node/slave-agent.jnlp
```
- Or if the slave is headless:

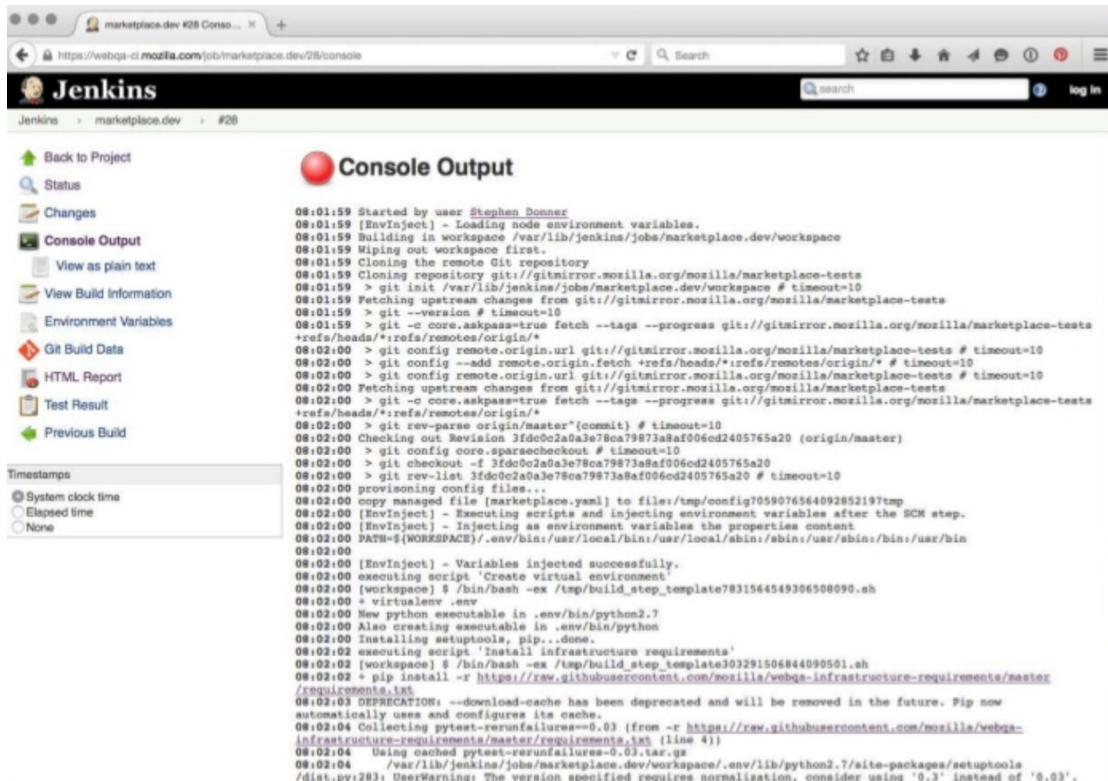
```
java -jar slave.jar -jnlpUrl http://localhost:8080/computer/Test%20Slave%20Node/slave-agent.jnlp
```

Created by anonymous user

Projects tied to Test Slave Node

None

Вихід на консоль завдань



The screenshot displays the Jenkins web interface in a browser window. The address bar shows the URL `https://webqa-ci.mozilla.com/job/marketplace.dev/28/console`. The Jenkins logo and navigation links are visible at the top. On the left sidebar, the 'Console Output' link is selected. The main panel, titled 'Console Output', shows a log of build steps and their execution details. The log includes timestamps, user information, and the output of various commands such as `git init`, `git fetch`, `git checkout`, and `pip install`. The build is identified as 'marketplace.dev #28' and was started by user 'Stephen Donner'.

marketplace.dev #28 Console...

https://webqa-ci.mozilla.com/job/marketplace.dev/28/console

Jenkins

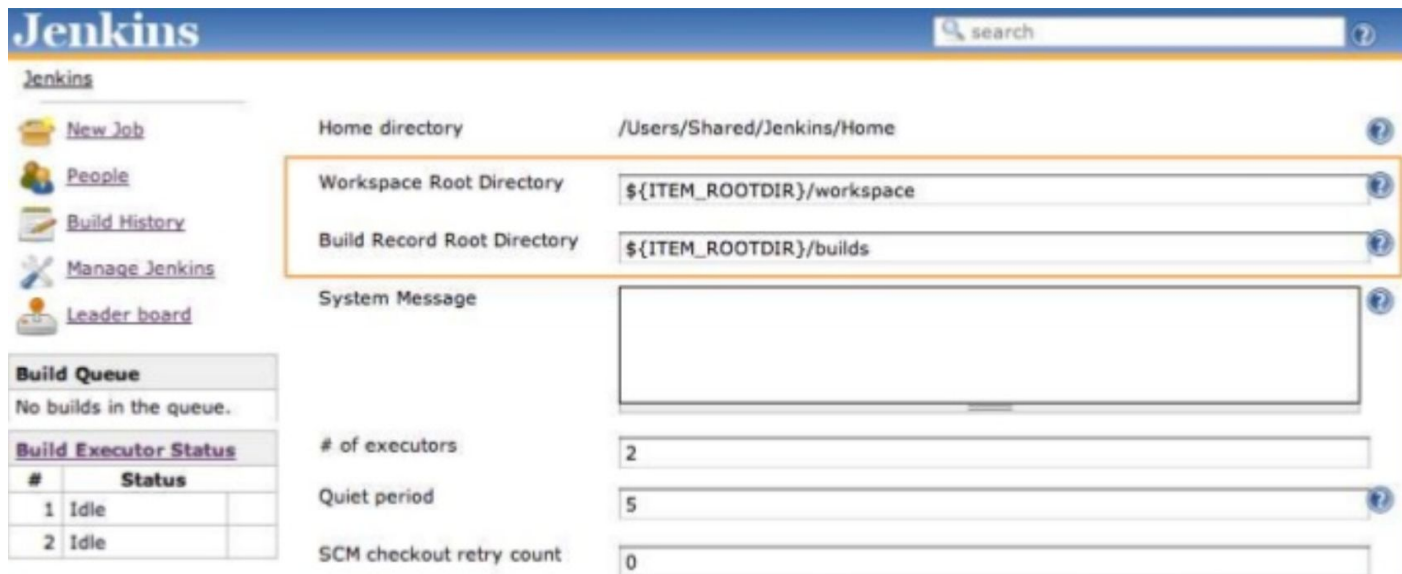
Back to Project
Status
Changes
Console Output
View as plain text
View Build Information
Environment Variables
Git Build Data
HTML Report
Test Result
Previous Build

Timestamps
System clock time
Elapsed time
None

Console Output

```
08:01:59 Started by user Stephen Donner
08:01:59 [EnvInject] - Loading node environment variables.
08:01:59 Building in workspace /var/lib/jenkins/jobs/marketplace.dev/workspace
08:01:59 Wiping out workspace first.
08:01:59 Cloning the remote Git repository
08:01:59 Cloning repository git://gitmirror.mozilla.org/mozilla/marketplace-tests
08:01:59 > git init /var/lib/jenkins/jobs/marketplace.dev/workspace # timeout=10
08:01:59 Fetching upstream changes from git://gitmirror.mozilla.org/mozilla/marketplace-tests
08:01:59 > git --version # timeout=10
08:01:59 > git -c core.askpass=true fetch --tags --progress git://gitmirror.mozilla.org/mozilla/marketplace-tests
+refs/heads/*:refs/remotes/origin/*
08:02:00 > git config remote.origin.url git://gitmirror.mozilla.org/mozilla/marketplace-tests # timeout=10
08:02:00 > git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
08:02:00 > git config remote.origin.url git://gitmirror.mozilla.org/mozilla/marketplace-tests # timeout=10
08:02:00 Fetching upstream changes from git://gitmirror.mozilla.org/mozilla/marketplace-tests
08:02:00 > git -c core.askpass=true fetch --tags --progress git://gitmirror.mozilla.org/mozilla/marketplace-tests
+refs/heads/*:refs/remotes/origin/*
08:02:00 > git rev-parse origin/master^{commit} # timeout=10
08:02:00 Checking out Revision 3fde0c2a0a3e78ca79873a8af006cd2405765a20 (origin/master)
08:02:00 > git config core.sparsecheckout # timeout=10
08:02:00 > git checkout -f 3fde0c2a0a3e78ca79873a8af006cd2405765a20
08:02:00 > git rev-list 3fde0c2a0a3e78ca79873a8af006cd2405765a20 # timeout=10
08:02:00 provisioning config files...
08:02:00 copy managed file [marketplace.yaml] to file:/tmp/config7059076564092852197tmp
08:02:00 [EnvInject] - Executing scripts and injecting environment variables after the SCM step.
08:02:00 [EnvInject] - Injecting as environment variables the properties content
08:02:00 PATH=${WORKSPACE}/.env/bin:/usr/local/bin:/usr/local/sbin:/bin:/usr/bin:/usr/bin
08:02:00 [EnvInject] - Variables injected successfully.
08:02:00 executing script 'Create virtual environment'
08:02:00 [workspace] $ /bin/bash -ex /tmp/build_step_template7831564549366508090.sh
08:02:00 + virtualenv .env
08:02:00 New python executable in .env/bin/python2.7
08:02:00 Also creating executable in .env/bin/python
08:02:00 Installing setuptools, pip...done.
08:02:02 executing script 'Install infrastructure requirements'
08:02:02 [workspace] $ /bin/bash -ex /tmp/build_step_template303291504844090501.sh
08:02:02 + pip install -r https://raw.githubusercontent.com/mozilla/webqa-infrastructure-requirements/master/requirements.txt
08:02:03 DEPRECATION: --download-cache has been deprecated and will be removed in the future. Pip now
automatically uses and configures its cache.
08:02:04 Collecting pytest-rerunfailures==0.03 (from -r https://raw.githubusercontent.com/mozilla/webqa-
infrastructure-requirements/master/requirements.txt (line 4))
08:02:04 Using cached pytest-rerunfailures-0.03.tar.gz
08:02:04 /var/lib/jenkins/jobs/marketplace.dev/workspace/.env/lib/python2.7/site-packages/setuptools
/dist.py:283: UserWarning: The version specified requires normalization, consider using '0.3' instead of '0.03'.
```

Кожна робота має свій власний простір



The screenshot shows the Jenkins configuration interface. On the left is a sidebar with navigation links: [Jenkins](#), [New Job](#), [People](#), [Build History](#), [Manage Jenkins](#), and [Leader board](#). Below these are sections for **Build Queue** (showing 'No builds in the queue.') and **Build Executor Status** (a table with 2 idle executors). The main area contains configuration fields: 'Home directory' (set to /Users/Shared/Jenkins/Home), 'Workspace Root Directory' (set to \${ITEM_ROOTDIR}/workspace), 'Build Record Root Directory' (set to \${ITEM_ROOTDIR}/builds), 'System Message' (a large text area), '# of executors' (set to 2), 'Quiet period' (set to 5), and 'SCM checkout retry count' (set to 0). An orange rectangular box highlights the 'Workspace Root Directory' and 'Build Record Root Directory' fields.

Jenkins search ?

[Jenkins](#)

[New Job](#)

[People](#)

[Build History](#)

[Manage Jenkins](#)

[Leader board](#)

Build Queue

No builds in the queue.

Build Executor Status

#	Status
1	Idle
2	Idle

Home directory: /Users/Shared/Jenkins/Home ?

Workspace Root Directory: \${ITEM_ROOTDIR}/workspace ?

Build Record Root Directory: \${ITEM_ROOTDIR}/builds ?

System Message: ?

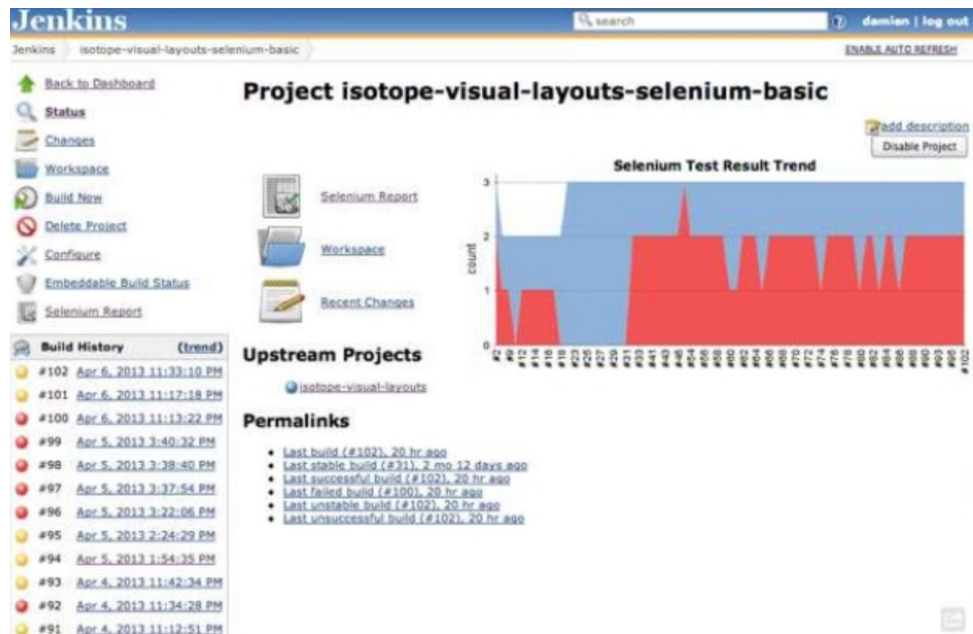
of executors: 2

Quiet period: 5 ?

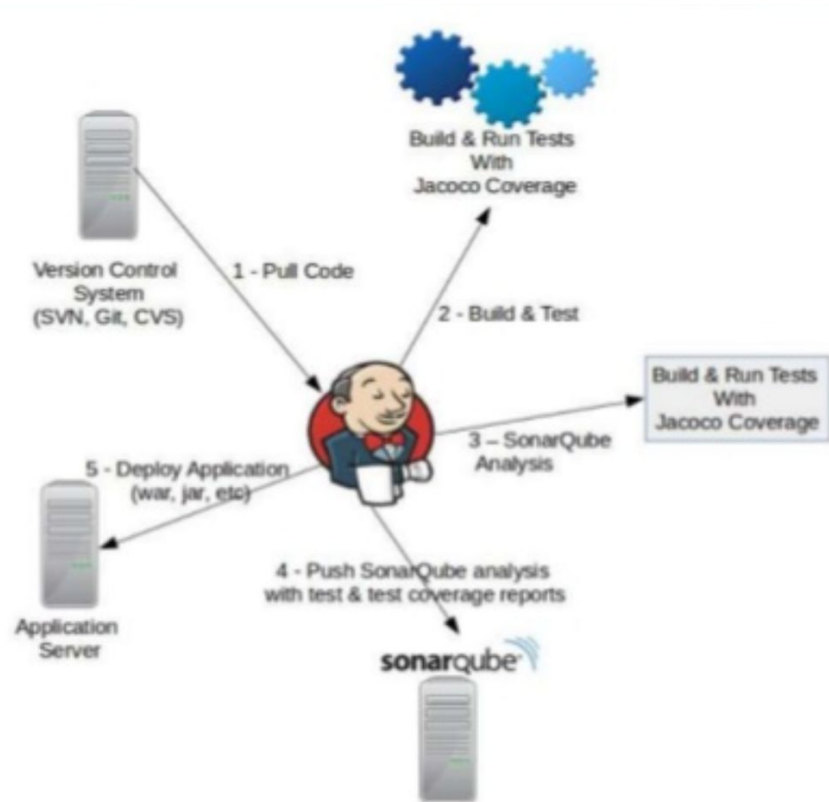
SCM checkout retry count: 0

Jenkins and Selenium Testing Tool (програмне забезпечення)

Добре працювати разом



Приклад того, як Дженкінс вмикає конвеєри CI/CD



Робота з багатокроковим кодом Python на Jenkins

Build

 **Execute Python script**

Script `execfile("jobscript.py")`

[See the list of available environment variables](#)

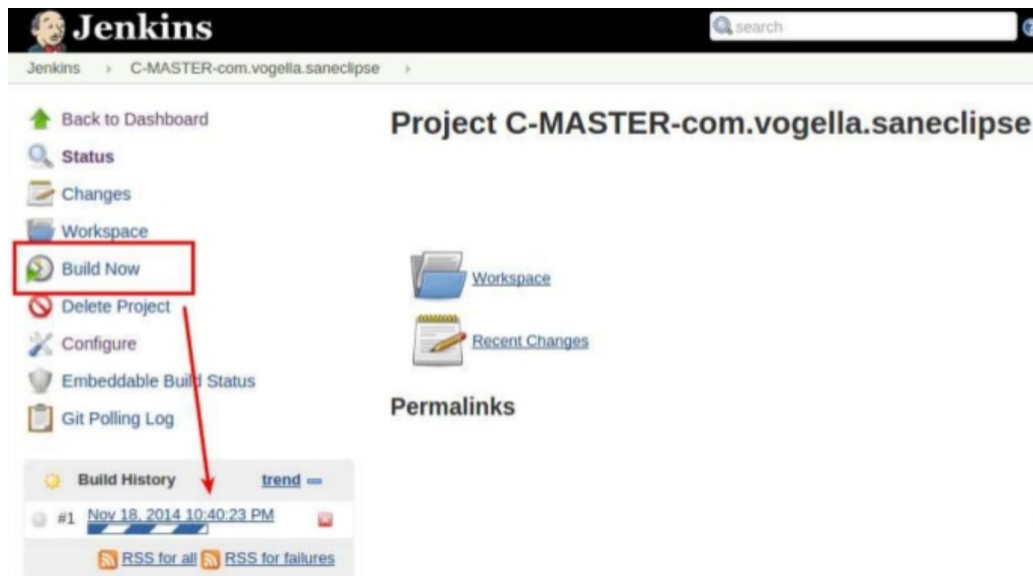
Add build step ▼

Post-build Actions

Add post-build action ▼

Save **Apply**

Розпочніть будівництво прямо зараз





Jenkins і безперервна інтеграція

Що це?

- Сервер безперервної інтеграції - виявляє зміни в Subversion, виконує завдання повторно. (Створення, тестування, розгортання, тестування, упаковка)
- Близько семи років
- 47000+ встановлень
- 600+ плагінів
- Відгалуження оригінального проекту Hudson

Навіщо нам це потрібно?

- Щоб *інтегрувати частіше*, швидше виявляти *помилки*, покращувати *якість* і зменшувати *витрати*
- Координуйте виконання завдань у рамках робочих процесів.
- Компілюйте, тестуйте та пакуйте, розгортайте, створюйте сценарії, перевіряйте, створюйте віртуальні машини

АЛЬТЕРНАТИВИ Jenkins

TeamCity – дозволяє створювати власні збірки, попередньо перевірені коміти

Bamboo — CI-сервер Atlassian, хороший, але дещо негнучкий

Cruise Control — застарілий, не відповідає сучасним вимогам

Cron і bash...

Термінологія

- Робота - одиниця роботи над проектом
- Перегляд – визначена користувачем колекція завдань або робочий процес
- Майстер - центральний майстер Дженкінса, виконує планування завдань
- Підлеглий вузол — виконує одну або кілька задач у слотах (виконавцях)
- Робоче місце - робоча зона, де виконується робота
- Плагін - визначена користувачем колекція завдань або робочий процес

Архітектура Jenkins



Головний інтерфейс користувача Jenkins

Jenkins search service / log out

Views

Build Queue
No builds in this queue.

Build Executor Status

#	Executor	Status
1	idle	Master
2	idle	Master
1	build1 (offline)	Master
	build2 (offline)	Master
	build3 (offline)	Master
	build4 (offline)	Master
	idle	Master

Icon	Name	Last Success	Last Failure	Last Duration	Start Job
🟢	glance	15 hr (#526)	5 days 15 hr (#527)	2 min 3 sec	🔴
🟢	glance-merge	15 hr (#119)	1 mo 2 days (#79)	3 sec	🔴
🟢	glance-pool	15 hr (#209)	16 days (#209)	6.5 sec	🔴
🟢	keystone	12 hr (#600)	13 hr (#599)	1 min 45 sec	🔴
🟢	keystone-merge	12 hr (#550)	6 days 15 hr (#567)	4.6 sec	🔴
🟢	keystone-pool	12 hr (#139)	6 days 15 hr (#135)	9 sec	🔴
🟢	keystone-verify	12 hr (#400)	6 days 15 hr (#393)	22 sec	🔴
🟢	nova	3 hr 44 min (#121729)	19 hr (#12171)	8 min 19 sec	🔴
🟢	nova-merge	3 hr 44 min (#215)	21 hr (#154)	1 min 0 sec	🔴
🟢	nova-pool	3 hr 44 min (#1588)	21 hr (#1567)	1 min 9 sec	🔴
🟢	quantum	9 hr 37 min (#50)	9 days 9 hr (#50)	9.1 sec	🔴
🟢	quantum-pool	9 hr 37 min (#50)	N/A	4.4 sec	🔴
🟡	quantum-verify	9 hr 35 min (#31)	N/A	20 sec	🔴
🟢	swift	21 hr (#119340)	1 mo 17 days (#119320)	14 sec	🔴
🟢	swift-merge	21 hr (#810)	N/A	7.3 sec	🔴

List of available jobs

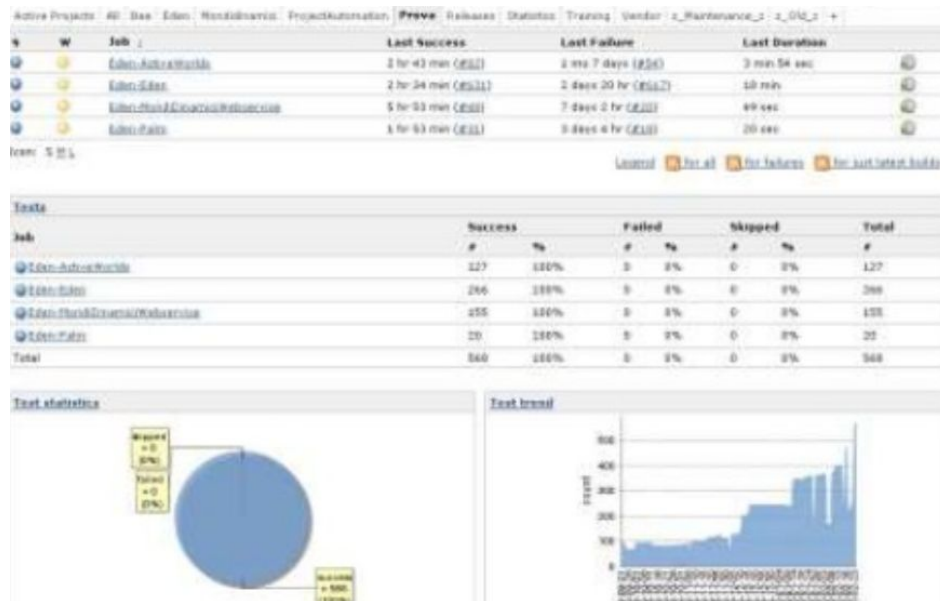
List of jobs being executed

Спеціальні режими перегляду Jenkins

Ви можете створювати власні
представлення для задач Jenkins

Задачі, які вас цікавлять і над якими ви
працюєте

Ви також можете створювати власні
представлення за допомогою плагіна
Dashboard



Дослідження причини збою

Детально подивіться журнал збірки

Спробуйте виконати завдання на підлеглому пристрої (через SSH)

Дозволи — поширена проблема

Плагіни Jenkins

- Subversion - опитування щодо змін до збірки
- Sonar - публікує метрики по збірках
- Parameterised triggers - тригер будується на умовах
- Copy Artifact - копіює артефакт для завершених збірок
- Робочий процес - налаштування робочих процесів побудови
- JClouds - положення середовищ

Jenkins Поради

Коли робота зазнає невдачі – заявіть про невдачу, щоб люди знали, що відбувається. Надішліть електронний лист

Використовуйте параметризовані збірки - не жорстко кодуйте шляхи до R, Java тощо... думайте про загальні

Не встановлюйте залежності на серверах збірки Jenkins. Створіть власне середовище!

Питання та відповіді